

1. Finding the Sum of Digits of a Number Until Zero

Difficulty: Easy

Topics: Basic Programming, Mathematical Computations

Description: Write a program to repeatedly sum the digits of a number until the result is zero.

Example:

Input: `number = 123`

Output: 6

Explanation: Sum of digits is $1 + 2 + 3 = 6$; sum of digits of 6 is 6 (which is a single digit).

2. Generating a Multiplication Table for a Range

Difficulty: Easy

Topics: Arrays, Basic Programming

Description: Write a program to generate multiplication tables for numbers within a specified range.

Example:

Input: `start = 2, end = 4`

Output:

```
2 x 1 = 2   3 x 1 = 3   4 x 1 = 4
2 x 2 = 4   3 x 2 = 6   4 x 2 = 8
2 x 3 = 6   3 x 3 = 9   4 x 3 = 12
2 x 4 = 8   3 x 4 = 12  4 x 4 = 16
```

3. Calculating the Sum of a Series ($1 + 1/2 + 1/3 + \dots + 1/n$)

Difficulty: Medium

Topics: Mathematical Computations

Description: Write a program to calculate the sum of the series $1 + 1/2 + 1/3 + \dots + 1/n$ up to the nth term.

Example:

Input: `n = 4`

Output: 2.083333

Explanation: Sum of the series is $1 + 1/2 + 1/3 + 1/4 \approx 2.083333$.

4. Finding All Pairs of Elements in an Array that Add Up to a Given Sum

Difficulty: Medium

Topics: Arrays, Hashing

Description: Write a program to find all pairs of elements in an array whose sum equals a specified target.

Example:

Input: `array = [1, 2, 3, 4, 5], target = 5`

Output: `[(1, 4), (2, 3)]`

Explanation: Pairs that sum up to 5 are (1, 4) and (2, 3).

5. Generating a Diamond Pattern of Stars

Difficulty: Medium

Topics: Patterns, Basic Programming

Description: Write a program to create a diamond pattern with stars of a given size.

Example:

Input: `size = 5`

Output:

```
  *
 ***
*****
 ***
  *
```

6. Counting the Number of Palindromic Substrings in a String

Difficulty: Medium

Topics: String Manipulation

Description: Write a program to count how many palindromic substrings exist in a given string.

Example:

Input: `string = "aaa"`

Output: 6

Explanation: Palindromic substrings are "a", "a", "a", "aa", "aa", "aaa".

7. Generating a Matrix with Multiplication Table

Difficulty: Easy

Topics: Arrays, Matrix Operations

Description: Write a program to generate a matrix where each element at position (i, j) is the product of i and j.

Example:

Input: `size = 3`

Output:

```
1 2 3
2 4 6
3 6 9
```

8. Finding the GCD of Multiple Numbers

Difficulty: Medium

Topics: Mathematical Computations

Description: Write a program to find the GCD (Greatest Common Divisor) of an array of numbers.

Example:

Input: `array = [12, 24, 36]`

Output: 12

Explanation: The GCD of 12, 24, and 36 is 12.

9. Finding the Sum of the First N Odd Numbers

Difficulty: Easy

Topics: Mathematical Computations

Description: Write a program to calculate the sum of the first N odd numbers.

Example:

Input: `N = 5`

Output: 25

Explanation: Sum of the first 5 odd numbers (1 + 3 + 5 + 7 + 9) is 25.

10. Finding the Number of Perfect Numbers Up to a Given Limit

Difficulty: Medium

Topics: Number Theory

Description: Write a program to find how many perfect numbers exist up to a given limit.

Example:

Input: `limit = 30`

Output: 1

Explanation: There is only one perfect number (6) up to 30.

11. Finding the Largest Prime Factor of a Number

Difficulty: Medium

Topics: Number Theory

Description: Write a program to find the largest prime factor of a given number.

Example:

Input: `number = 28`

Output: 7

Explanation: The prime factors of 28 are 2 and 7, with the largest being 7.

12. Generating a Matrix of Fibonacci Numbers

Difficulty: Medium

Topics: Arrays, Matrix Operations

Description: Write a program to generate a matrix where each element is a Fibonacci number.

Example:

Input: `size = 3`

Output:

```
1 1 2
3 5 8
13 21 34
```

13. Finding the Sum of the First N Prime Numbers

Difficulty: Medium

Topics: Prime Numbers, Mathematical Computations

Description: Write a program to calculate the sum of the first N prime numbers.

Example:

Input: `N = 4`

Output: 17

Explanation: The sum of the first 4 prime numbers (2 + 3 + 5 + 7) is 17.

14. Checking for a Balanced Bracket Sequence

Difficulty: Medium

Topics: String Manipulation, Stack

Description: Write a program to check if a given string with brackets is balanced.

Example:

Input: `string = "{[()]}"`

Output: `True`

Explanation: The brackets are balanced.

15. Finding the Sum of Numbers in a String

Difficulty: Easy

Topics: String Manipulation

Description: Write a program to extract and sum all numbers present in a given string.

Example:

Input: `string = "The numbers are 12 and 34"`

Output: `46`

Explanation: The sum of numbers 12 and 34 is 46.

16. Finding the Longest Consecutive Sequence in an Array

Difficulty: Medium

Topics: Arrays, Hashing

Description: Write a program to find the longest sequence of consecutive numbers in an array.

Example:

Input: `array = [100, 4, 200, 1, 3, 2]`

Output: `4`

Explanation: The longest consecutive sequence is [1, 2, 3, 4].

17. Generating a Matrix with a Spiral Pattern

Difficulty: Medium

Topics: Arrays, Matrix Operations

Description: Write a program to generate a matrix filled with numbers in a spiral pattern.

Example:

Input: `size = 3`

Output:

```
1 2 3
8 9 4
7 6 5
```

18. Finding All Subsets of a Set

Difficulty: Medium

Topics: Combinatorics

Description: Write a program to generate all possible subsets of a given set of numbers.

Example:

Input: `set = [1, 2]`

Output: `[[], [1], [2], [1, 2]]`

Explanation: The subsets of `[1, 2]` are the empty set, `[1]`, `[2]`, and `[1, 2]`.

19. Checking for Perfect Squares in a Range

Difficulty: Easy

Topics: Mathematical Computations

Description: Write a program to check which numbers in a given range are perfect squares.

Example:

Input: `start = 1, end = 10`

Output: `[1, 4, 9]`

Explanation: Perfect squares between 1 and 10 are 1, 4, and 9.

20. Finding the Sum of Diagonal Elements in a Matrix

Difficulty: Easy

Topics: Matrix Operations

Description: Write a program to calculate the sum of the diagonal elements in a square matrix.

Example:

Input: `matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`

Output: 15

Explanation: The sum of the diagonal elements (1 + 5 + 9) is 15.

21. Finding the Second Smallest Number in an Array

Difficulty: Easy

Topics: Arrays

Description: Write a program to find the second smallest number in an array.

Example:

Input: `array = [12, 13, 1, 10, 34, 1]`

Output: 10

Explanation: The second smallest number in the array is 10.

22. Generating Pascal's Triangle Up to N Rows

Difficulty: Medium

Topics: Combinatorics

Description: Write a program to generate Pascal's Triangle up to N rows.

Example:

Input: `N = 3`

Output:

```
1
1 1
1 2 1
```

23. Finding the Sum of Digits of the Product of Two Numbers

Difficulty: Easy

Topics: Mathematical Computations

Description: Write a program to find the sum of the digits of the product of two given numbers.

Example:

Input: `number1 = 12, number2 = 34`

Output: 9

Explanation: The product of 12 and 34 is 408, and the sum of its digits is $4 + 0 + 8 = 12$.

24. Checking for Palindromic Numbers in a Range

Difficulty: Medium

Topics: Mathematical Computations

Description: Write a program to check for palindromic numbers within a given range.

Example:

Input: `start = 1, end = 100`

Output: [1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 22, 33, 44, 55, 66, 77, 88, 99]

Explanation: Palindromic numbers between 1 and 100 are the numbers listed.

25. Generating a Matrix with Alternating 0s and 1s

Difficulty: Easy

Topics: Arrays, Matrix Operations

Description: Write a program to generate a matrix where the elements alternate between 0 and 1.

Example:

Input: `size = 3`

Output:

```
0 1 0
1 0 1
0 1 0
```

26. Finding the Count of a Specific Word in a String

Difficulty: Easy

Topics: String Manipulation

Description: Write a program to count how many times a specific word appears in a given string.

Example:

Input: `string = "hello world hello"`

Output: 2

Explanation: The word "hello" appears 2 times in the string.

27. Finding the Largest Sum of a Subarray

Difficulty: Medium

Topics: Arrays, Dynamic Programming

Description: Write a program to find the largest sum of any contiguous subarray.

Example:

Input: array = [-2, 1, -3, 4, -1, 2, 1, -5, 4]

Output: 6

Explanation: The largest sum is 6, from the subarray [4, -1, 2, 1].

28. Generating a Right-Angle Triangle Pattern of Numbers

Difficulty: Easy

Topics: Patterns

Description: Write a program to create a right-angle triangle pattern with numbers.

Example:

Input: height = 4

Output:

```
1
12
123
1234
```

29. Finding All Divisors of the Product of Two Numbers

Difficulty: Medium

Topics: Number Theory

Description: Write a program to find all divisors of the product of two given numbers.

Example:

Input: number1 = 6, number2 = 10

Output: [1, 2, 3, 5, 6, 10, 15, 30]

Explanation: The product of 6 and 10 is 60, and its divisors are listed.

30. Finding the Longest Sequence of Consecutive 1s in a Binary Array

Difficulty: Medium

Topics: Arrays, Binary Operations

Description: Write a program to find the longest sequence of consecutive 1s in a binary array.

Example:

Input: array = [1, 1, 0, 1, 1, 1]

Output: 3

Explanation: The longest sequence of 1s is [1, 1, 1] with length 3.

31. Calculating the Sum of the First N Fibonacci Numbers

Difficulty: Medium

Topics: Fibonacci Sequence, Mathematical Computations

Description: Write a program to calculate the sum of the first N Fibonacci numbers.

Example:

Input: N = 5

Output: 12

Explanation: The first 5 Fibonacci numbers are 1, 1, 2, 3, 5, and their sum is 12.

32. Checking for a Repeated Substring in a String

Difficulty: Medium

Topics: String Manipulation

Description: Write a program to check if a substring is repeated within a given string.

Example:

Input: `string = "abab"`

Output: `True`

Explanation: The substring "ab" is repeated.

33. Finding the Median of a List of Numbers

Difficulty: Medium

Topics: Sorting, Mathematical Computations

Description: Write a program to find the median value of a list of numbers.

Example:

Input: `list = [3, 1, 4, 1, 5]`

Output: `3`

Explanation: After sorting the list to `[1, 1, 3, 4, 5]`, the median is 3.

34. Finding the Number of Words in a String

Difficulty: Easy

Topics: String Manipulation

Description: Write a program to count the number of words in a given string.

Example:

Input: `string = "Hello world"`

Output: `2`

Explanation: There are 2 words in the string.

35. Generating a Matrix with a Diagonal Pattern

Difficulty: Medium

Topics: Matrix Operations

Description: Write a program to create a matrix where elements form diagonal lines of a given pattern.

Example:

Input: `size = 4`

Output:

```
1 0 0 0
1 1 0 0
1 1 1 0
1 1 1 1
```

36. Finding the Sum of the First N Even Numbers

Difficulty: Easy

Topics: Mathematical Computations

Description: Write a program to calculate the sum of the first N even numbers.

Example:

Input: $N = 4$

Output: 20

Explanation: The first 4 even numbers are 2, 4, 6, 8, and their sum is 20.

37. Finding the Count of Digits Greater Than a Specific Value

Difficulty: Easy

Topics: Mathematical Computations

Description: Write a program to count how many digits in a number are greater than a specific value.

Example:

Input: `number = 54321, value = 3`

Output: 2

Explanation: The digits

greater than 3 in 54321 are 5, 4, and 4, so the count is 2.

38. Generating a Pattern of Prime Numbers

Difficulty: Medium

Topics: Prime Numbers, Patterns

Description: Write a program to generate a pattern where each row contains the first few prime numbers.

Example:

Input: `rows = 3`

Output:

```
2
2 3
2 3 5
```

39. Finding the Common Elements in Two Arrays

Difficulty: Medium

Topics: Arrays

Description: Write a program to find common elements between two arrays.

Example:

Input: `array1 = [1, 2, 3, 4], array2 = [3, 4, 5, 6]`

Output: `[3, 4]`

Explanation: The common elements between the two arrays are 3 and 4.

40. Finding the Sum of the Squares of All Even Numbers Up to N

Difficulty: Medium

Topics: Mathematical Computations

Description: Write a program to calculate the sum of squares of all even numbers up to a given N.

Example:

Input: `N = 4`

Output: 20

Explanation: The even numbers up to 4 are 2 and 4, and their squares are 4 and 16. The sum is 20.

41. Generating a Pattern of Increasing Numbers

Difficulty: Easy

Topics: Patterns

Description: Write a program to create a pattern where numbers increase with each row.

Example:

Input: `rows = 3`

Output:

```
1
12
123
```

42. Finding the Largest Element in Each Row of a Matrix

Difficulty: Easy

Topics: Matrix Operations

Description: Write a program to find the largest element in each row of a matrix.

Example:

Input: `matrix = [[1, 2, 3], [4, 5, 6], [7, 8, 9]]`

Output: `[3, 6, 9]`

Explanation: The largest elements in each row are 3, 6, and 9.

43. Checking for Anagram Pairs in a List of Strings

Difficulty: Medium

Topics: String Manipulation

Description: Write a program to find pairs of strings in a list that are anagrams of each other.

Example:

Input: `strings = ["listen", "silent", "hello", "world"]`

Output: `[("listen", "silent")]`

Explanation: "listen" and "silent" are anagrams.

44. Finding the Frequency of Each Character in a String

Difficulty: Easy

Topics: String Manipulation

Description: Write a program to count the frequency of each character in a given string.

Example:

Input: `string = "hello"`

Output: `{'h': 1, 'e': 1, 'l': 2, 'o': 1}`

Explanation: The frequency of each character in the string "hello" is shown.

45. Generating a Matrix with Random Numbers

Difficulty: Easy

Topics: Random Number Generation, Matrix Operations

Description: Write a program to generate a matrix filled with random numbers.

Example:

Input: `rows = 2, columns = 3`

Output:

```
3 8 1
7 4 6
```

46. Finding the Length of the Longest Word in a String

Difficulty: Easy

Topics: String Manipulation

Description: Write a program to find the length of the longest word in a given string.

Example:

Input: `string = "Find the longest word"`

Output: 8

Explanation: The longest word is "longest" with length 8.

47. Finding All Triplets in an Array That Sum to Zero

Difficulty: Medium

Topics: Arrays, Sorting

Description: Write a program to find all unique triplets in an array that sum to zero.

Example:

Input: `array = [-1, 0, 1, 2, -1, -4]`

Output: `[[-1, -1, 2], [-1, 0, 1]]`

Explanation: The unique triplets that sum to zero are listed.

48. Generating a Square Matrix with Random Values

Difficulty: Easy

Topics: Random Number Generation, Matrix Operations

Description: Write a program to generate a square matrix where each element is a random value.

Example:

Input: `size = 3`

Output:

```
7 3 5
2 6 9
1 8 4
```

49. Finding the Difference Between the Sum of Even and Odd Numbers in an Array

Difficulty: Easy

Topics: Arrays, Mathematical Computations

Description: Write a program to calculate the difference between the sum of even and odd numbers in an array.

Example:

Input: `array = [1, 2, 3, 4, 5, 6]`

Output: 4

Explanation: The sum of even numbers is 12, and the sum of odd numbers is 8. The difference is 4.

50. Generating a Triangle Pattern of Stars with a Given Height

Difficulty: Easy

Topics: Patterns

Description: Write a program to create a triangle pattern of stars with a specified height.

Example:

Input: `height = 4`

Output:

```
*  
* *  
* * *  
* * * *
```